

Evaluation of nonlinear static analysis methods and software tools for seismic analysis of highway bridges

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Abstract

The response of bridges when subjected to seismic excitation can be evaluated by a number of analysis methods. The traditional approach is to employ linear static or dynamic analyses coupled with appropriate modifiers to account for inelastic response while current design practice is moving towards an increased emphasis on nonlinear static analysis methods. In this study, a preliminary seismic response analysis of a two-span highway bridge was performed using linear dynamic analysis procedures to identify the potential for inelastic response. The bridge was subsequently analyzed in accordance with two nonlinear static analysis methods (capacity spectrum and inelastic demand spectrum methods) in an effort to evaluate the difference in global response predicted by the two methods and the effectiveness of various software programs for performing nonlinear static analysis. The results demonstrated that, for the two nonlinear static analysis methods used in this study, different predictions of nonlinear response occurred with neither method being regarded as producing correct results due to a number of simplifications inherent in the methods. Furthermore, the analysis revealed that some software programs are well suited to performing nonlinear static analysis, both from the point-of-view of performing comprehensive analyses and providing a graphical depiction of the evolution of damage (i.e. the behaviour of plastic hinges).

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1. Introduction

The seismic design of typical highway bridges in the United States is governed by the American Association of Highway and Transportation Officials bridge design specifications [1]. The specifications permit the designer to utilize a variety of methods for seismic analysis, from simple equivalent static analysis to complex nonlinear dynamic analysis. For simple bridges, it is common practice to utilize methods that employ linear analysis, accounting for nonlinear response through the application of force reduction factors. This approach has several shortcomings, which have been accepted due to its simplicity and a lack of alternative practical approaches [21]. Such an approach may be regarded as *force-based* since the primary

emphasis of the methods is on the forces within the structure. In addition to a linear analysis, it is common practice to perform a capacity analysis associated with the desired inelastic response in which ductile flexural response occurs at selected plastic hinge regions within the structure. The plastic hinge regions are detailed to ensure plastic behaviour while inhibiting nonductile failure modes.

Since damage potential and ultimate failure can usually be directly related to the inelastic displacement capacity of the structure [8], in recent years there has been a shift of attention away from linear methods of seismic analyses to nonlinear methods which put emphasis on the displacements within the structure. Thus, nonlinear methods of analysis that are capable of realistically predicting the deformations imposed by earthquakes on structures are needed. In response to this need, new nonlinear static analysis procedures have appeared in national resource documents such as the ATC-40 report [4] on seismic evaluation and retrofit of

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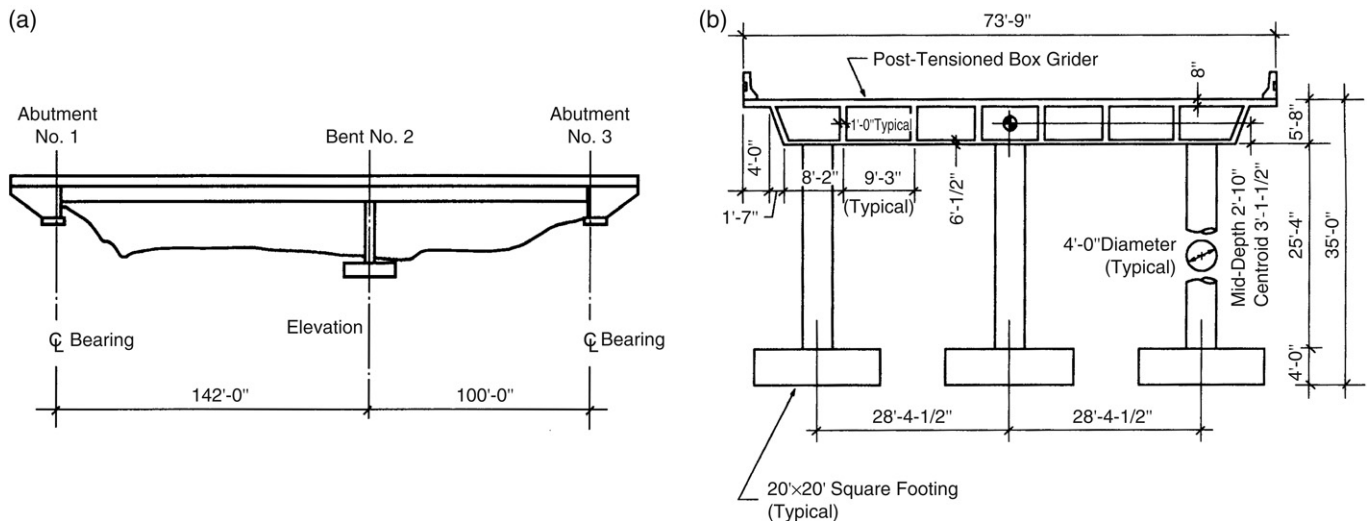


Fig. 1. Elevation and cross-section of bridge (from [28]) (1 in. = 2.54 cm).

concrete buildings and the FEMA-356 prestandard on seismic rehabilitation of buildings [16]. Such methods of analysis are useful for predicting inelastic displacement capacities while simultaneously offering a compromise between the oversimplification of linear static analysis and the inherent complexity of nonlinear dynamic analysis.

The *Capacity Spectrum Method*, originally developed by Freeman [18,17], is one of the more frequently utilized methods of nonlinear static analysis and is the principal method described in the ATC-40 report [4]. As described by Yu et al. [34], the Capacity Spectrum Method begins with a nonlinear static pushover analysis which results in a graphical depiction of the global force–displacement relation for the structure. Note that pushover analysis procedures are described in a number of references (e.g. see [24,22]). The demand on the structure is then represented graphically by elastic spectra with equivalent viscous damping. However, a suitable value for the equivalent viscous damping is difficult to ascertain since a stable relationship between the hysteretic energy dissipation associated with the maximum excursion and equivalent viscous damping does not exist [23]. Several methods have been proposed to overcome the deficiencies of the original version of the Capacity Spectrum Method. For example, the *Displacement Coefficient Method* [16] represents demand via inelastic displacement spectra which are obtained from the elastic displacement spectra using a number of correction factors, which in principle are expected to be more accurate than elastic spectra with equivalent viscous damping [14]. Bertero [6] recommended the use of smoothed inelastic design response spectra to represent the demand whereas Chopra and Goel [10] developed the *Capacity-Demand-Diagram Method* that represents demand via a constant-ductility design spectrum. Efforts continue to be made to improve upon current forms of nonlinear static analysis methods and to incorporate such methods in design documents [15,5].

Due to the growing interest in evaluating the seismic performance of bridges via nonlinear static analysis methods,

there is a need to assess the suitability of various analysis methods and software tools for performing practical seismic analysis of structures. This study addresses that need, particularly as it applies to common (two- or three-span) highway bridge structures, by examining the seismic response of a two-span highway bridge using two nonlinear static analysis methods and various software tools.

2. Description of bridge structure and finite element model

The highway bridge under study is taken from Design Example No. 1 of the Federal Highway Administration (FHWA) Seismic Design of Bridges Series [27] and was designed in accordance with the Single-Mode Spectral Method of the 15th Edition of the *AASHTO Standard Specifications for Highway Bridges*, Division I, as amended by the Interim Specifications-Bridges-1993 through 1995 [2]. The bridge consists of a two-span continuous, post-tensioned, reinforced concrete box girder with a three-column integral bent, and spread footings (see Fig. 1) and was selected for investigation based on its similarity to typical highway bridges in the state of Washington.

Two different support configurations were considered. In the *Basic Support Configuration*, the bridge has seat-type abutments which allow limited longitudinal movement of the superstructure due to the gap between the superstructure and the abutment back wall. The support provided by the abutment is assumed to be fixed against translation in the vertical and transverse directions and fixed against rotation about the longitudinal axis while the column bent footings are considered to be fixed against both translation and rotation.

In the *Spring Support Configuration*, the bridge has stub wall abutments which are restrained in the longitudinal and transverse directions due to end diaphragm and wing wall interaction with the soil, respectively. The support provided by the abutment is assumed to be fixed against translation vertically, fixed against rotation about the longitudinal axis of the superstructure and has translational springs in the

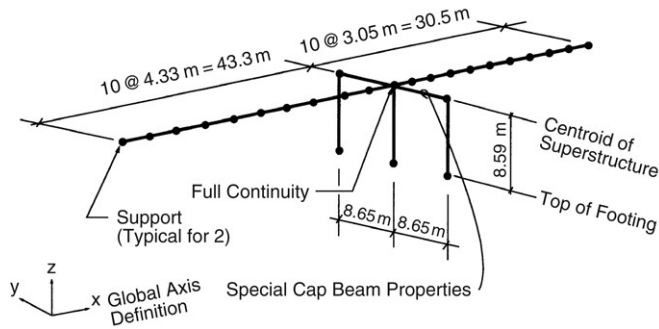


Fig. 2. Finite element model of bridge (adapted from [28]).

longitudinal and transverse directions while the column bent footings were supported on translational and rotational springs in each orthogonal direction to account for soil flexibility. Note that, for the column bent footings, the support configuration presented in [27] does not include the translational springs or the vertical rotational spring that are included in this study. For the abutment supports, the stiffness of the translational springs was determined in accordance with the Caltrans bridge design specifications [7] wherein the soil profile is assumed to consist of 250 ft of glacial sand and gravel. For the bent footings, spring-only foundations were used [29] with the stiffness of the translational and rotational springs being determined based on two studies [19,25] which are presented in the prestandard on seismic rehabilitation of buildings [16]. A comprehensive discussion on the modelling of the footings and the abutments can be found in [33].

The finite element model of the bridge (see Fig. 2) is represented by three-dimensional frame elements which pass through the mid-depth of the superstructure and three-dimensional frame elements which pass through the geometrical centre and mid-depth of the columns and the cap beam, respectively. A single frame element was deemed sufficient for the columns since they are relatively short and prismatic. Since the cross-section of the superstructure is uniform, it was deemed sufficient to locate nodes at the tenth points of each span. The mass of each member was equally distributed to its end nodes. The seismic weight of the bridge is 21.7 MN which includes half of the bent weight. The superstructure cross-sectional area is 11.15 m² and the second moment of area about the strong and weak cross-sectional axes are 440.2 m⁴ and 4.96 m⁴, respectively. The box-girder is assumed to be integral with the bent and thus full continuity is employed at the superstructure-bent intersection. The torsional rigidity about the longitudinal axis of the elements representing the superstructure is increased by several orders of magnitude over the actual value to reflect the physical torsional restraint due to the spatial support of the superstructure on the bent and abutments. The cap beam cross-sectional area is 2.32 m² and the second moment of area about the strong and weak cross-sectional axes and the torsional rigidity about the longitudinal axis were increased by several orders of magnitude over the actual values to ensure that, as would expected for the actual bridge, the columns attract approximately equal forces. The circular columns had a diameter of 1.22 m and a longitudinal

reinforcement ratio of 1.9%. As per standard practice, the bridge columns were designed to preclude shear failures in favor of flexural failures. A rigid end zone (0.863 m in length) is located at the top of the elements representing the columns to account for the offset between the centreline of the cap beam and the soffit of the box girder. It is assumed that the concrete has a nominal 28-day compressive strength of 27.6 kPa and a modulus of elasticity of 24,831 kPa and that the transverse and longitudinal reinforcement has a nominal yield strength of 413.9 kPa.

3. Description of seismic analysis procedures

In this study, preliminary seismic analysis of the bridge was performed using linear dynamic analysis procedures in the form of response spectrum and response-history methods. Such methods are advantageous over linear static methods in that they can provide insight into the dynamics of the bridge due to their ability to explicitly account for the effects of multiple modes of vibration. Furthermore, the results of linear dynamic analysis can be used to determine whether significant inelastic behaviour is likely to occur and thus can be used to select cases for which more complex static or dynamic nonlinear analysis is warranted. Linear dynamic analysis methods may be regarded as *force-based* since the emphasis is on forces rather than displacements. In fact, due to inelastic behavior, it is unlikely that the displacements will be properly estimated in such analyses.

The inelastic seismic response of the bridge was evaluated via two nonlinear static analysis methods (Capacity Spectrum Analysis and Inelastic Demand Spectrum Analysis). Such methods may be regarded as *displacement based* since the emphasis is on displacements rather than forces. A brief explanation of the procedure for conducting nonlinear static analysis is as follows. After applying gravity loads to the bridge, a pseudostatic pushover analysis of the bridge is performed to establish the *pushover curve* which is then transformed to a *capacity curve* which characterizes the bridge response in its fundamental mode of vibration. During the pushover analysis, a lateral force distribution or displacement pattern (e.g. proportional to the fundamental mode shape) is applied in an incremental fashion while monitoring the occurrence of nonlinear behaviour and the displacement of a control node (e.g. the centre of mass of the bridge superstructure). The pushover curve is established as the base shear (V_b) versus control node displacement (u_N) and has an endpoint associated with a predefined displacement limit or the ultimate capacity of the bridge. The ultimate capacity may correspond to either a localized failure (e.g. a single plastic hinge reaching its curvature capacity) or the development of a global collapse mechanism. The pushover curve is then converted to a capacity curve for an equivalent single-degree of freedom (SDOF) system by converting the base shear to the fundamental mode pseudo-spectral acceleration (A) and the control node displacement to the fundamental mode spectral displacement (D) via the following transformations which are obtained from

basic principles of modal analysis (see, for example, [9]):

$$A = \frac{V_b}{M_1^*} \quad (1)$$

$$D = \frac{u_N}{\phi_{N1}\Gamma_1} \quad (2)$$

where M_1^* is the effective modal mass for the fundamental mode, ϕ_{N1} is the modal amplitude of the control node in the fundamental mode, and Γ_1 is the modal participation factor for the fundamental mode.

Modal analysis inherently involves the superposition of modal responses and thus, strictly speaking, only applies to linear systems. Since inelastic static analysis procedures begin with the development of a nonlinear pushover curve, it is evident that the aforementioned transformation equations are not strictly valid. To address this discrepancy, alternative nonlinear static analysis procedures have been developed. For example, the ATC-40 report [4] indicates that the transformation equations can be modified by updating the fundamental mode shape to reflect the effects of nonlinear response. Additional alternatives to the inelastic static analysis procedure described above include the application of a fundamental mode loading distribution that adapts to changes in the nonlinear response of the structure (e.g. see [13]) and the consideration of response in multiple modes of vibration [4,30].

In the *Capacity Spectrum Analysis* method, the demand is expressed in the form of elastic SDOF Acceleration-Displacement Response Spectra (ADRS) corresponding to various viscous damping ratios. The seismic performance of the structure is then assessed by overlaying the capacity and demand curves, recognizing that each point on the capacity curve can be associated with an equivalent viscous damping ratio and natural period. The point at which the capacity curve intersects a demand curve associated with the same viscous damping ratio is the performance point which defines the spectral displacement demand. The control node displacement and base shear demand for the actual multidegree of freedom (MDOF) structure are then obtained from Eqs. (1) and (2).

The equivalent viscous damping ratio of the inelastic system, ξ_{eq} , is obtained by equating the energy dissipated in one cycle of motion for the inelastic system and the equivalent linear system. For bilinear systems with strain-hardening, the equivalent viscous damping ratio can be determined as follows [10,4]:

$$\xi_{eq} = \frac{2(\mu - 1)(1 - \alpha)}{\pi\mu(1 + \alpha\mu - \alpha)} \quad (3)$$

where μ is the displacement ductility of the structure and α is the ratio of post to preyielding stiffness. The total equivalent viscous damping ratio, ξ , is equal to the inherent viscous damping [assumed to be 5% as per ATC-40] plus the equivalent viscous damping from Eq. (3). In addition, the equivalent viscous damping may be modified by a damping modification factor, κ , which accounts for the deviation of the actual hysteresis loops from the idealized parallelogram associated with bilinear hysteresis [4]. In this study, the value of κ is taken as its largest possible value (i.e. unity).

The equivalent natural period, T_{eq} , of the bilinear system is based on the secant stiffness at a given displacement (ductility) level and is given by [10]:

$$T_{eq} = T_n \sqrt{\frac{\mu}{1 + \alpha\mu - \alpha}} \quad (4)$$

where T_n is the natural period associated with elastic behavior.

The second nonlinear static analysis method used in this study was the *Inelastic Demand Spectrum Analysis* method which is essentially the same as the improved *Capacity-Demand-Diagram Method* proposed by Chopra and Goel [10]. The method is similar to the Capacity Spectrum Analysis method except that the demand curves, rather than being represented by equivalent inelastic response spectra, are represented by actual inelastic response spectra. In this study, to maintain consistency with the linear dynamic analyses in which the demand corresponded to selected historical earthquake records, the seismic demand is represented by inelastic response spectra corresponding to each of the earthquake records. The inelastic response spectra were developed in the form of pseudo-yield acceleration response spectra with the assumption of elasto-plastic behaviour and considering a range of ductility levels. The demand curves were then obtained by converting the yield acceleration spectra to the ADRS domain. The seismic performance of the bridge was assessed by overlaying the capacity and demand curves. The point at which the capacity curve intersects a demand curve associated with the same level of ductility is the performance point which defines the spectral displacement demand. The control node displacement and base shear demand for the actual MDOF structure are then obtained from Eqs. (1) and (2).

4. Analysis software and seismic excitation

The seismic response of the bridge was determined using the following software: GT-STRUDL version 25 [20], SAP2000 Nonlinear version 7.0 [11], ADINA 800-node version [3], and SC-Push3D version 2.0 [31]. The software was selected based a survey of bridge engineering consulting firms and State DOT's [33] which was conducted in an attempt to identify the most commonly used analysis methods and associated software for seismic analysis of highway bridges. In a separate survey conducted as part of the ATC-55 Project [15], the most common software used by practicing engineers for nonlinear static analysis of buildings was SAP2000. Note that ADINA was not used for the inelastic static analyses since this program provided no convenient tools for conducting such analyses. The seismic excitation used in the analyses was represented by the following historical earthquake records: (1) Weak Event—N86E component of the Olympia (Washington DOT Highway Test Lab) record of the 1949 Western Washington Earthquake; (2) Moderate Event—S00E component of the El Centro record of the 1940 Imperial Valley, California Earthquake; and (3) Strong Event—N90E component of the Sylmar Hospital Parking Lot record of the 1994 Northridge, California Earthquake. The rationale for selecting these three records was as follows: the Olympia record corresponds to

a relatively strong earthquake in the geographical region of interest (i.e. western Washington state), the El Centro record has spectral characteristics that are known to closely match the design spectral shapes adopted in the Uniform Building Code on firm soil sites [8], and the Sylmar record is from the Northridge Earthquake, an earthquake which caused the partial or complete collapse of five bridges while damaging approximately 200 others [12]. Note that seismic response analyses using a small number of records was deemed sufficient for this study since the records were adequate for demonstrating the difference in global performance that can be predicted via the two methods of nonlinear static analysis.

5. Linear dynamic analysis of bridge

A linear elastic model of the bridge was developed for the purpose of performing eigenvalue analysis (to identify natural periods and mode shapes) and for performing preliminary seismic analysis to identify the potential for inelastic response. The seismic analysis was performed for both the Basic and Spring Supported bridge models and with excitation applied nonconcurrently in the transverse and longitudinal directions. Two linear dynamic analysis procedures were utilized (response spectrum and response-history analysis) with damping in the bridge being characterized by an assumed 5% modal damping for each mode of vibration. The eigenvalue analysis resulted in the longitudinal and transverse fundamental natural periods and associated modal participating mass ratios (i.e. effective modal mass to total mass ratios) shown in Table 1. Due to the significant difference in the natural periods for the Basic and Spring Support Conditions in the longitudinal direction (see Table 1), the elastic seismic response was quite different for these two cases with inelastic response expected only for the Basic Support Conditions. More specifically, the results from the seismic analyses (i.e. the moments and axial forces in the columns due to combined dead and seismic loads) revealed that inelastic behaviour of the columns is likely to occur for the moderate and strong events (El Centro and Sylmar Records) in the longitudinal direction (for transverse excitation the abutments resist most of the force) for the Basic Support Conditions [33]. Consequently, nonlinear static analysis was performed for the aforementioned conditions. In addition, only the fundamental mode of vibration (longitudinal motion) was considered in the nonlinear static analyses since the natural period of the fundamental mode (0.72 s) is less than one second [4] and the modal participating mass ratio for the fundamental mode was 91.9% which is greater than the typical 90% minimum participating mass to include in seismic analyses. Finally, in performing the analysis in the longitudinal direction, it was assumed that the bent resists all of the inertial force generated within the bridge deck (i.e., any resistance provide by the abutments is neglected).

6. Development of pushover and capacity curves

6.1. Pushover curve from SC-Push3D

The pushover analysis in SC-Push3D assumes that inelastic response occurs in concentrated plastic hinges located at the

Table 1

Summary of fundamental natural periods and associated participating mass ratios for longitudinal and transverse directions

		Basic support model	Spring support model
Natural period (s)	Transverse	0.170	0.271
	Longitudinal	0.716	0.187
Modal participating mass ratios (%)	Transverse	85.6	95.9
	Longitudinal	91.9	97.7

ends of each element. For the bridge model used in this study, the potential plastic hinges were restricted to the columns, while the elements representing the superstructure were assumed to remain elastic. The behavior of the plastic hinges is characterized by a yield surface and a rigid-elastic/plastic moment–rotation relation. The plastic hinge length is defined as an equivalent length over which the curvature is considered to be constant and equal to the curvature at the critical (end) section of the member. The moment-curvature relation for the columns of the bridge (see Fig. 3) was obtained using the program SEQMC [32] wherein Mander's confined concrete model is used [26]. The idealized bilinear moment-curvature relation shown in Fig. 3 was obtained using the following anchor points: (1) the theoretical first yield in the reinforcement; (2) the ideal section moment capacity (i.e. the idealized yield point); and (3) the ultimate limit state (i.e. the point at which the confined compression strain is equal to 2.2%).

The pushover curve was obtained by performing a displacement-controlled analysis wherein the bridge was subjected to a prescribed displacement pattern. The use of displacement-controlled analysis does not require that the displacement be defined at all nodes; rather, the displacement may be applied at only a few nodes that dominate the response (e.g. nodes that dominate the fundamental mode response). In this study, a displacement was applied in the longitudinal direction at the control node located at the intersection of the top of the centre column and the superstructure (see Fig. 2). The displacement was applied in a monotonically increasing fashion until a specified level of displacement was achieved. The resulting base shear versus control node displacement represents the pushover curve.

During application of the control node displacement, either local failure or global collapse may occur. A local failure is associated with reaching the curvature capacity at a hinge (resulting in a hinge that cannot be relied upon to provide continued resistance to reversed cyclic loading). Note that the bridge may be globally stable under the local failure and, in this case, the failed element must be manually removed from the bridge model before proceeding (i.e. SC-Push3D provides notification that the ultimate flexural strength at a hinge is reached but does not automatically remove the element). The occurrence of a global collapse condition (i.e. forming sufficient plastic hinges to produce a mechanism) is not

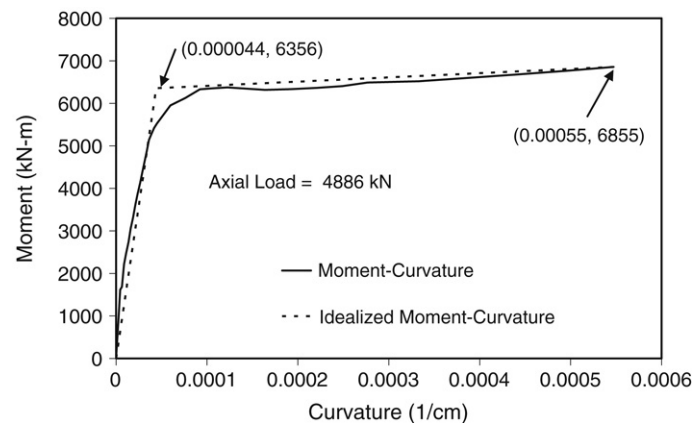


Fig. 3. Moment–curvature relationship for columns of bridge model.

monitored by SC-Push3D and thus the user must check for this possibility.

The pushover curve obtained from the analysis using SC-Push3D is shown in Fig. 4. The initial negative displacement is the result of application of dead loads prior to the pushover analysis. The pushover analysis results in plastic hinges developing at the ends of all three columns and local failure occurring at the top of the centre column at a displacement of 5.94 cm. Note that the post-yield stiffness is quite small since all of the plastic hinges are initiated at approximately the same displacement.

6.2. Pushover curve from SAP2000

Pushover analysis in SAP2000 assumes that nonlinear behaviour occurs within frame elements at concentrated plastic hinges with default or user-defined hinge properties being assigned to each hinge. In this study, a coupled axial force and biaxial bending moment hinge (P-M2-M3 hinge) was assigned to the ends of the columns of the bridge model. The default hinge properties are section-dependent and are typically based on the nonlinear modelling parameters given in ATC-40 [4]. Note that, although the ATC-40 nonlinear modelling parameters are characterized by performance levels associated with building behaviour (i.e. Immediate Occupancy, Life Safety, and Collapse Prevention), the performance levels are useful for interpreting the pushover response of bridge structures. The hinge properties are defined through the definition of the moment–rotation relation, the interaction surface, and acceptance criteria. In this study, the moment–rotation relation was determined using SAP2000 default parameter values (yield rotation automatically determined based on the defined material properties and post-yield stiffness equal to 5% of elastic stiffness). A user-defined interaction surface was developed by defining a number of equally spaced axial force–moment curves between 0 and 360 degrees. For the P-M2-M3 hinge, the acceptance criteria deformations are plastic rotations normalized by the yield rotation for immediate occupancy (IO), life safety (LS) and collapse prevention (CP) performance levels.

The pushover curve was obtained by first analysing the bridge under the effect of dead load and then pushing the bridge

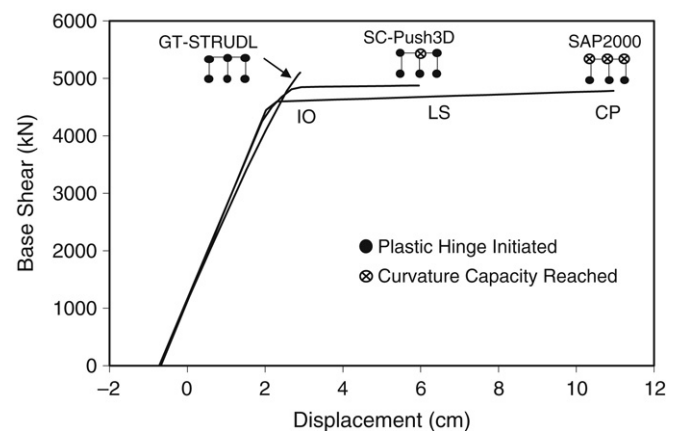


Fig. 4. Pushover curves from analysis of bridge in the longitudinal direction and with basic support conditions.

longitudinally using the uniform acceleration load pattern (a predefined load pattern in SAP2000) until either a prescribed maximum displacement was achieved at the control node or the bridge failed. Note that SAP2000 automatically stops the analysis when a plastic hinge reaches its curvature capacity (by default, defined to be a plastic hinge rotation equal to six times the yield rotation), requiring the user to check if a sufficient number of plastic hinges have developed to render the structure globally unstable. The pushover analysis results in plastic hinges developing at the ends of all three columns and the CP performance level being reached at the top of the columns at a displacement of 10.9 cm (see Fig. 4). At this point, the hinges at the top of the columns are regarded as having reached their curvature capacity and the analysis is stopped. Note that, at each stage of loading, SAP2000 employs colour-codes to define hinge behaviour which allows the user to observe the evolution of damage in the structure.

6.3. Pushover curve from GT-STRUDL

The pushover analysis in GT-STRUDL assumes that nonlinear behaviour occurs at plastic hinges located at the ends of members. A distributed plasticity model is used to characterize the plastic hinge behaviour wherein the hinge region is subdivided into concrete and steel fibers. The hinge

behaviour is defined by the stress–strain relation of the fibres, the geometry of the cross-section and the assumed length of the plastic hinge. In this study, the compressive stress–strain behaviour of the confined concrete was based on the stress–strain model presented by Mander et al. [26].

The pushover analysis was performed by first applying the bridge dead load and then applying a series of incremental uniform traction loads along the longitudinal axis of the superstructure (a convenient tool for applying a displacement pattern in lieu of a load pattern is not available). In GT-Strudl, failure is defined as the development of sufficient plastic hinges to establish a potential failure mechanism, even though the hinges may have additional deformation capacity. The pushover analysis results in plastic hinges developing at the ends of all three columns with global instability being detected at a displacement of 2.9 cm (see Fig. 4). At this point, 84% of the plastic hinge fibres have yielded at the top and bottom of the outboard columns and 73% at the top and bottom of the centre column.

6.4. Comparison of pushover curves

From Fig. 4, it is evident that the three software programs produce essentially identical results for the initial linear dead load analysis and the elastic portion of the pushover analysis. However, due to the difference in how failure is defined by each program, the results are significantly different for the nonlinear portion of the pushover analysis. For the bridge under study, all three software programs indicate that a global collapse mechanism occurs at a control node displacement of approximately 2.5 cm. However, the bridge is capable of continuing to carry lateral load, with SAP2000 indicating the displacement capacity is approximately 10.9 cm, at which time the plastic hinges at the top of all three columns reach their curvature capacity. A re-run of the SC-Push analysis wherein the definition of failure was revised to be consistent with the default definition used by SAP2000 (i.e. an ultimate rotation of six times the yield rotation) results in a failure displacement that is approximately equal to that obtained with SAP2000. Note that, due to the difference in the plastic hinge model (GT-STRUDL employs a distributed plasticity model whereas SC-Push3D and SAP2000 employ a concentrated plasticity model), for displacements between about 0.65 and 2.5 cm, the pushover curve obtained from GT-STRUDL is distinctly different from the curves obtained from SC-Push3D and SAP2000.

6.5. Development of capacity curve

Of the four software programs used in this study, SAP2000 is the only program that is capable of generating the capacity curve and demand curves for seismic performance evaluation. Thus, from this point forward, the results from SAP2000 are emphasized. Applying the transformations defined by Eqs. (1) and (2) to the pushover curve of Fig. 4, and removing the initial negative displacement due to the dead load effect, leads to the capacity curve shown in Fig. 5. Note that, in SAP2000, the initial point on the capacity curve is determined based on

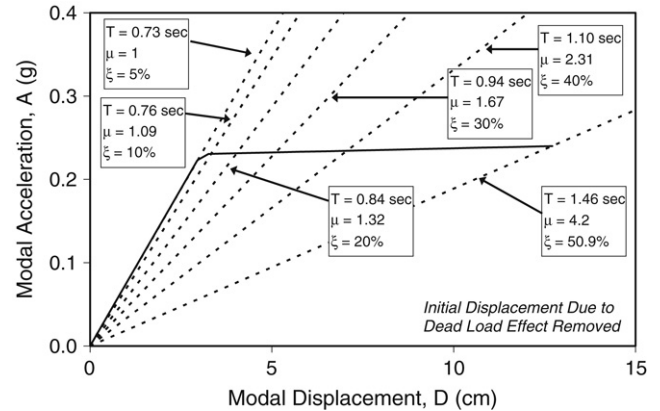


Fig. 5. Capacity curve with effective period, ductility and total damping ratio identified at selected displacements.

the magnitude of the initial displacement (i.e. for the bridge under study, the initial point on the capacity curve generated by SAP2000 has a positive displacement). However, it is suggested herein that a more appropriate approach is to retain the sign of the initial negative displacement and then remove the dead load effect before performing the capacity spectrum analysis. In the Capacity Spectrum Method, the demand curve is in the form of an ADRS curve in which lines of constant natural period radiate out from the origin. In order to compare the capacity curve with the ADRS demand curve, the displacement due to dead load must be removed from the capacity curve so that the lines of constant effective natural period on the capacity curve coincide with the associated lines on the ADRS demand curve. Thus, for purposes of seismic performance evaluation, the lateral displacement due to dead load is temporarily removed and will be subsequently accounted for in the seismic performance evaluation.

To evaluate the seismic performance of the bridge, it is necessary to determine the ductility and total viscous damping at various points along the capacity curve. These properties were determined by replacing the capacity curve with a simplified bilinear curve. Since the capacity curve is nearly bilinear (see Fig. 5), the bilinear curve was generated by extending the pre- and post-yielding regions of the capacity curve to find their common intersection point which was then defined as the yield point for the bilinear capacity curve. From the bilinear capacity curve, the post- to pre-yielding stiffness ratio and the ductility level at selected displacements is determined. Finally, the total viscous damping ratio at the selected displacements is obtained by adding the 5% inherent damping to the equivalent viscous damping ratio defined by Eq. (3). The results are shown graphically in Fig. 5 where lines of constant effective natural period, determined using Eq. (4), radiate out from the origin. Note that the ductility levels and damping ratios shown in Fig. 5 are associated with the points on the capacity curve that intersect the radial lines.

7. Development of demand curves

In the Capacity Spectrum Analysis method, the seismic demand is represented by a series of ADRS demand curves for

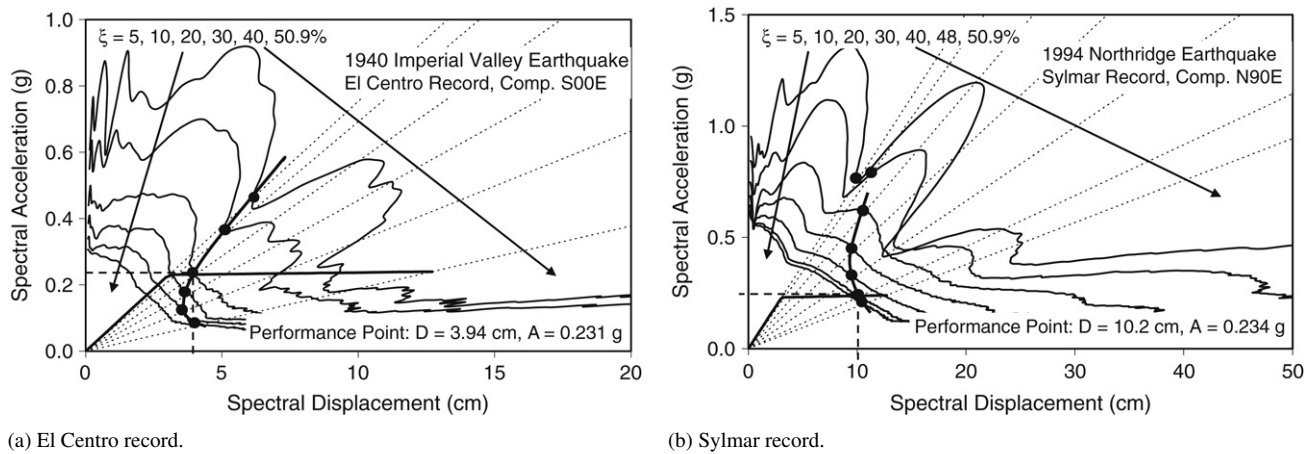


Fig. 6. Graphical performance evaluation of bridge using capacity spectrum analysis.

elastic SDOF systems. The ADRS demand curves for damping ratios ranging from 5% to 50.9% are shown in Fig. 6 for the El Centro and Sylmar earthquake records. The lowest damping level, 5%, corresponds to inherent damping associated with elastic behaviour whereas the maximum damping ratio, 50.9%, corresponds to combined elastic and inelastic behaviour at the Collapse Prevention performance level (see Fig. 5).

The Inelastic Demand Spectrum method represents the seismic demand using inelastic ADRS spectrums for SDOF systems. In this study, the inelastic response spectra were generated assuming elastoplastic behaviour and using standard procedures (e.g. see [9]). The ADRS spectra for 5%-damped elastoplastic systems having a ductility ranging from 1 to 8 and subjected to the El Centro and Sylmar earthquake records are shown in Fig. 7 for a natural period range of 0.02–3 s. Due to the time-consuming process of generating inelastic response spectra, the natural period increment is not constant and may be somewhat large (up to 0.5 s), giving rise to the somewhat jagged appearance of the spectra.

8. Inelastic static analysis of bridge

8.1. Capacity Spectrum Analysis

According to the Capacity Spectrum Analysis method, the seismic performance of the bridge can be evaluated by overlaying the capacity curve with the elastic ADRS demand curves (see Fig. 6). The procedure utilized in this study for identifying the performance point is a graphical procedure that is essentially the same as that described as Procedure C in ATC-40 [4]. The procedure may be described as follows: For each viscous damping ratio, identify the point of intersection between the ADRS demand curve for that damping ratio and the radial line of constant period on the capacity curve associated with that damping ratio. The locus of such points has been indicated by solid circles in Fig. 6. The performance point is that point which coincides with the capacity curve. As shown in Fig. 6(a), for the El Centro record, the performance point is associated with a total viscous damping ratio of 20%. For the Sylmar record (see Fig. 6(b)), the performance point did

Table 2

Summary of bridge response from inelastic static analyses

	Capacity Spectrum Analysis			Inelastic Demand Spectrum Analysis		
	u_N (cm)	V_b (kN)	Performance	u_N (cm)	V_b (kN)	Performance
El Centro	2.92	4604.4	IO	5.72	4604.4	LS
Sylmar	8.61	4664.5	LS/CP	–	–	Collapse

not lie on one of the radial lines associated with a preselected damping ratio and thus trial damping ratios were employed until the performance point coincided with the radial line associated with a total viscous damping ratio of 48%.

The performance of the bridge is evaluated based on its response as a MDOF structural system. The performance points identified in Fig. 6 are associated with a SDOF response of the bridge and the associated MDOF response can be determined from Eqs. (1) and (2). The final results are given in Table 2 with due consideration given to the initial negative displacement due to dead loads. A comparison of the control node displacements shown in Table 2 with the pushover curve shown in Fig. 4 indicates that, for the El Centro record, the bridge is predicted to just achieve the Immediate Occupancy (IO) performance level while, for the Sylmar record, the bridge performance is predicted to be approximately midway between the Life Safety (LS) and Collapse Prevention (CP) performance level.

8.2. Inelastic Demand Spectrum Analysis

According to the Inelastic Demand Spectrum Analysis method, the seismic performance of the bridge can be evaluated by overlaying the capacity curve with the inelastic ADRS demand curves (see Fig. 7). The procedure utilized to identify the performance point parallels that described previously for the Capacity Spectrum method except that the performance point is associated with a certain level of ductility rather than a viscous damping ratio. The performance point for each ground motion is indicated in Fig. 7. The performance of the bridge based on its response as a MDOF structural system is shown in Table 2.

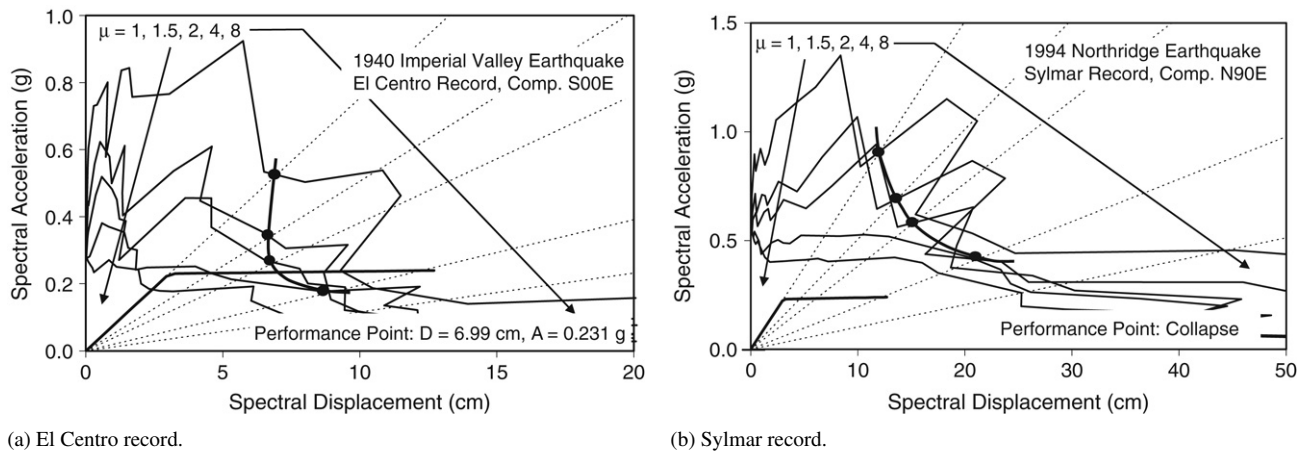


Fig. 7. Graphical performance evaluation of bridge using inelastic demand spectrum analysis.

A comparison of the control node displacements shown in Table 2 with the pushover curve shown in Fig. 4 indicates that, for the El Centro record, the bridge is predicted to just achieve the Life Safety (LS) performance level while, for the Sylmar record, the bridge is predicted to collapse.

As an alternative to generating inelastic ADRS curves, the so-called *Inelastic Static Analysis* method may be employed in which only the yield acceleration response spectrum is utilized [34]. In this case, an assessment of performance is made by using the yield acceleration response spectrum to determine the yield acceleration demand associated with the fundamental natural period and the ductility capacity of the bridge; the ductility capacity being taken from an idealized bilinear representation of the capacity curve. The maximum displacement demand is then obtained by relating the yield acceleration to the yield displacement. A comparison of the yield acceleration and displacement demands to the corresponding capacities from the capacity curve (at the point of maximum ductility) reveals whether the bridge will fail or not. For the bridge discussed in this study, this method indicated that the bridge would survive the El Centro record but fail under the Sylmar record which is the same conclusion regarding the survivability of the bridge as that obtained from the Inelastic Demand Spectrum Analysis. In fact, of the two inelastic static analysis methods evaluated in this study, the Inelastic Demand Spectrum Analysis method may be regarded as being more accurate than the Capacity Spectrum Analysis method since it makes use of actual inelastic demand spectra rather than equivalent inelastic spectra. On the other hand, as mentioned previously, the process of generating inelastic response spectra for specific earthquake records with high resolution (small period steps) is time-consuming and thus lower resolution (and thus lower accuracy) response spectra may be required from a practical point of view.

9. Discussion

The eigenvalue analysis and linear dynamic analysis performed herein revealed that the dynamic behaviour of the bridge was significantly influenced by the support conditions.

The influence of support conditions was particularly strong for seismic excitation in the longitudinal direction where the different abutment types resulted in significant differences in lateral resistance. The results from the linear dynamic analyses were used to identify cases (i.e. bridge configuration and/or seismic loading) under which significant inelastic response was expected. For the bridge under study, strong inelastic response was expected for the Basic Support Conditions and seismic excitation (El Centro and Sylmar records) in the longitudinal direction. Thus, nonlinear static analysis was performed only for this case.

The pushover analysis results were dependent on the assumptions used in the computer models, particularly the approach used to define the plastic hinges (e.g. concentrated versus distributed plasticity models) and the method by which failure was defined. Failure can be defined either by a plastic hinge reaching its curvature capacity (local failure) or by a sufficient number of plastic hinges being developed such that the structure is rendered globally unstable (global failure). In one program, both local failures and global failure must be checked by the user while in another local failure is automatically checked but not global failure. The third program defined failure as the development of sufficient plastic hinges to establish a potential failure mechanism, even though the hinges may have additional deformation capacity.

Only one of the software programs was capable of performing a comprehensive nonlinear static analysis wherein the capacity curve is generated and is overlaid with demand curves to evaluate the performance of the structure and a graphical depiction of the evolution of damage is provided. Interestingly, for the bridge studied herein, the software-generated capacity curve did not appear to correctly account for the initial displacement produced by the gravity loads. Further investigation of this issue is warranted.

The two inelastic static analysis procedures employed in this study require the development of demand curves in the form of ADRS spectra. In the Capacity Spectrum Analysis method, the demand is represented by elastic response spectra for a range of viscous damping ratios while, for the Inelastic Demand Spectrum method, the demand is represented by inelastic

response spectra for a range of ductility levels. In the case of *design* demand spectra, the effort required to generate the elastic and inelastic spectra is essentially the same. However, in the case of demand spectra for specific earthquake records, the effort required to generate inelastic demand spectra is quite significant and is regarded as a significant drawback to the Inelastic Demand Spectrum method.

Both the Capacity Spectrum Analysis method and the Inelastic Demand Spectrum analysis method were used to evaluate the performance of the bridge. The results from the two methods were different, with the Inelastic Demand Spectrum Analysis method predicting larger displacement demands on the bridge to the extent that lower global performance levels were predicted with this method. Although the Inelastic Demand Spectrum Analysis method may be regarded as being more accurate than the Capacity Spectrum Analysis method since it makes use of actual inelastic demand spectra rather than equivalent inelastic spectra, neither method of analysis can be regarded as producing correct results since a number of important issues are not explicitly considered in nonlinear static analyses (e.g. duration and cyclic nature of loading). Nonetheless, inelastic static analysis methods continue to appeal to practicing structural engineers due to their explicit emphasis on the post-yield response of structures and the graphical nature of some of the methods. The loss of accuracy is accepted since the methods are regarded as providing reasonable estimates of the inelastic demands on the structure, particularly with respect to the estimates provided by linear analysis procedures.

10. Conclusions

The conclusions of the study may be summarized as follows:

- The need for performing nonlinear seismic analysis is influenced by both the bridge configuration and seismic loading. Linear analysis methods may be conveniently used to prioritize cases under which nonlinear analysis should be conducted.
- Pushover analysis can be performed using a variety of software programs with the resultant pushover curve being dependent on the approach used to define the plastic hinge behaviour and the method by which failure is defined. Some programs are particularly well-suited to pushover analysis. However, of those programs evaluated herein, only one program offered a comprehensive nonlinear static analysis in that it included utilities for generating the pushover curve (with the ability to graphically track the evolution of damage), capacity curve and demand curves, identifying the SDOF performance point and converting back to the MDOF domain.
- Regarding the nonlinear static analysis methods used in this study, the Capacity Spectrum Analysis method is well-known and the seismic demand can be readily determined. For the Inelastic Demand Spectrum method, the seismic demand is much more difficult to define unless a simplified inelastic design spectrum is utilized.

- The approach to converting the pushover curve to the capacity curve when gravity loads produce initial displacements needs further investigation. Herein, it is suggested that the performance point should be identified in the absence of any dead load effect and the influence of dead loads accounted for in the transformation from the SDOF modal domain back to the MDOF physical domain.
- For the bridge structure examined herein, the Capacity Spectrum Analysis method and the Inelastic Demand Spectrum Analysis method led to different predictions of displacement demand. However, neither method is regarded as producing correct results due to a number of simplifications inherent in the methods. In spite of such shortcomings, inelastic static analysis methods remain attractive to practicing engineers due to their explicit consideration of inelastic response and due to the graphical nature of the seismic performance evaluation.

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